

**SAFETY DATA SHEET**

Creation Date 22-September-2009

Revision Date 26-March-2024

Revision Number 3

**1. Identification**

**Product Name** Calcium shot

**Cat No. :** 10127

**CAS-No** 7440-70-2  
**Synonyms** Calcium Metal.; Calciat

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

Substances/mixtures which, in contact with water, emit  
flammable gases

Category 2

Gas(es) = Hydrogen

**Label Elements****Signal Word**

Danger

**Hazard Statements**

In contact with water releases flammable gas

**Precautionary Statements****Prevention**

Do not allow contact with water

Wear protective gloves/protective clothing/eye protection/face protection

Handle and store contents under inert gas. Protect from moisture

**Response**

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

**Storage**

Store in a dry place. Store in a closed container

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component | CAS-No    | Weight % |
|-----------|-----------|----------|
| Calcium   | 7440-70-2 | 99       |

### 4. First-aid measures

|                                                               |                                                                                                                                                |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention if symptoms occur.              |
| <b>Skin Contact</b>                                           | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention if symptoms occur. |
| <b>Inhalation</b>                                             | Remove from exposure, lie down. Remove to fresh air.                                                                                           |
| <b>Ingestion</b>                                              | Clean mouth with water. Get medical attention.                                                                                                 |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                                                          |

### 5. Fire-fighting measures

|                                         |                                                                 |
|-----------------------------------------|-----------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                        |
| <b>Flash Point</b>                      | No information available                                        |
| <b>Method -</b>                         | No information available                                        |
| <b>Autoignition Temperature</b>         | No information available                                        |
| <b>Explosion Limits</b>                 |                                                                 |
| <b>Upper</b>                            | No data available                                               |
| <b>Lower</b>                            | No data available                                               |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                        |
| <b>Sensitivity to Static Discharge</b>  | No information available                                        |

**Specific Hazards Arising from the Chemical**

Water reactive. Produce flammable gases on contact with water.

**Hazardous Combustion Products**

Hydrogen. Calcium oxides.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

Health  
1

Flammability  
2

Instability  
0

Physical hazards  
W

**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Do not expose spill to water. Do not let this chemical enter the environment.

**7. Handling and storage****Handling**

Do not allow contact with water. Avoid contact with skin and eyes. Avoid contact with skin and clothing. Do not ingest. If swallowed then seek immediate medical assistance. Protect from moisture.

**Storage.**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from moisture. Keep from any possible contact with water. Incompatible Materials. Acids. Strong oxidizing agents. Alcohols. Ammonia. Halogens. oxygen.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures**

None under normal use conditions.

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No special protective equipment required.

**Recommended Filter type:** Particulates filter conforming to EN 143

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                        |                                |
|----------------------------------------|--------------------------------|
| Physical State                         | Solid                          |
| Appearance                             | Grey                           |
| Odor                                   | Odorless                       |
| Odor Threshold                         | No information available       |
| pH                                     | 14                             |
| Melting Point/Range                    | 845 °C / 1553 °F               |
| Boiling Point/Range                    | 1484 °C / 2703.2 °F @ 760 mmHg |
| Flash Point                            | No information available       |
| Evaporation Rate                       | Not applicable                 |
| Flammability (solid,gas)               | No information available       |
| Flammability or explosive limits       |                                |
| Upper                                  | No data available              |
| Lower                                  | No data available              |
| Vapor Pressure                         | No information available       |
| Vapor Density                          | Not applicable                 |
| Specific Gravity                       | No information available       |
| Solubility                             | Reacts violently with water    |
| Partition coefficient; n-octanol/water | No data available              |
| Autoignition Temperature               | No information available       |
| Decomposition Temperature              | No information available       |
| Viscosity                              | Not applicable                 |
| Molecular Formula                      | Ca                             |
| Molecular Weight                       | 40.07                          |

**10. Stability and reactivity**

|                                  |                                                                     |
|----------------------------------|---------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                 |
| Stability                        | Moisture sensitive.                                                 |
| Conditions to Avoid              | Incompatible products. Exposure to moist air or water.              |
| Incompatible Materials           | Acids, Strong oxidizing agents, Alcohols, Ammonia, Halogens, oxygen |
| Hazardous Decomposition Products | Hydrogen, Calcium oxides                                            |
| Hazardous Polymerization         | No information available.                                           |
| Hazardous Reactions              | None under normal processing.                                       |

**11. Toxicological information****Acute Toxicity**

|                             |                                                             |
|-----------------------------|-------------------------------------------------------------|
| Product Information         | No acute toxicity information is available for this product |
| Component Information       |                                                             |
| Toxicologically Synergistic | No information available                                    |

**Products****Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------|-----------|------------|------------|------------|------------|------------|
| Calcium   | 7440-70-2 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

Reacts with water so no ecotoxicity data for the substance is available. Do not empty into drains.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT**

|                      |         |
|----------------------|---------|
| UN-No                | UN1401  |
| Proper Shipping Name | CALCIUM |
| Hazard Class         | 4.3     |
| Packing Group        | II      |

**TDG**

|                      |         |
|----------------------|---------|
| UN-No                | UN1401  |
| Proper Shipping Name | CALCIUM |
| Hazard Class         | 4.3     |
| Packing Group        | II      |

**IATA**

|       |        |
|-------|--------|
| UN-No | UN1401 |
|-------|--------|

|                      |         |
|----------------------|---------|
| Proper Shipping Name | CALCIUM |
| Hazard Class         | 4.3     |
| Packing Group        | II      |
| <b>IMDG/IMO</b>      |         |
| UN-No                | UN1401  |
| Proper Shipping Name | CALCIUM |
| Hazard Class         | 4.3     |
| Packing Group        | II      |

## 15. Regulatory information

### International Inventories

| Component | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Calcium   | 7440-70-2 | X   | -    | X    | ACTIVE                                        | 231-179-5 | -      | -   |

| Component | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------|-----------|-------|----------|------|------|------|------|-------|-------|
| Calcium   | 7440-70-2 | X     | KE-04462 | X    | -    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

Authorisation/Restrictions according to EU REACH Not applicable

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Calcium   | 7440-70-2 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Calcium   | 7440-70-2 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

Prepared By Product Safety Department

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|                         |                                                    |
|-------------------------|----------------------------------------------------|
| <b>Creation Date</b>    | 22-September-2009                                  |
| <b>Revision Date</b>    | 26-March-2024                                      |
| <b>Print Date</b>       | 26-March-2024                                      |
| <b>Revision Summary</b> | New emergency telephone response service provider. |

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**